

i have one more deliverables which is data analysis for visualization
this will feed HMI back end --> then HMI back end feed --> front end for visualization
this is different from time series delivery---> this is for detecting any process anomaly
apart from process anomaly we need to visualize all KPIs, tcp poses (overlay on 3D geometry)
is also required
from 5.1 (data acquisition we will get cleaned, filtered and validated data)

can you give list of tasks associated with this?
also days or weeks required to complete the tasks
i am looking for detailed information about the tasks

Yes. Based on your latest scope, this should actually be a **separate deliverable** from:

- Time-Series Anomaly Detection (WP5.3 Analytics)
- Data-Driven Behavioral Model (System Identification)
- HMI Backend
- HMI Frontend

because this layer acts as the **bridge between process data and visualization**.

In other words:

5.1 Data Acquisition



Validated / Cleaned Data



Dx Visualization Analytics Layer



HMI Backend



HMI Frontend

This deliverable is responsible for preparing and transforming process data into visualization-ready information.

New Deliverable

Dx – Visualization Analytics & Process Mapping Framework

Objective

Transform process data, KPI data, machine states, TCP trajectories, anomaly outputs, and behavioral-model outputs into visualization-ready information that can be consumed by the HMI.

This layer does not perform anomaly detection.

This layer does not perform system identification.

It prepares data for visualization.

Expected Outputs

- KPI visualization datasets
 - Process-state visualization datasets
 - TCP trajectory visualization datasets
 - 3D geometry overlay datasets
 - Expected-vs-Actual visualization datasets
 - Anomaly visualization datasets
 - Root-cause visualization datasets
 - HMI-ready APIs and payloads
-

Total Estimated Duration

8–10 weeks

Dx.1 KPI Visualization Data Framework

Duration

1.5 weeks

Detailed Task

The first activity is identifying all KPIs that must appear inside the HMI.

The team will review outputs from:

- 5.1 Data Acquisition
- Time-Series Analysis
- System Identification

and determine:

- which KPIs need visualization
- update frequency
- aggregation level
- historical retention

Examples:

Tension

Pressure

Temperature

Speed

Machine Health

Layup Stability

Process Quality Indicators

For each KPI the visualization structure must be defined:

Current Value

Trend

Min

Max

Moving Average

Status

Output

- KPI catalog
 - KPI visualization schema
-

Dx.2 Process-State & Event Visualization Framework

Duration

1 week

Detailed Task

The objective is converting raw machine states and process events into meaningful visualization objects.

Examples:

Machine Running
Machine Idle
Process Start
Process Stop
Alarm Event
Pause Event

The framework should define:

State transitions
State durations
Timeline representation
Color coding
Severity categories

These outputs will later appear in process timelines inside HMI.

Output

- Process-state model
- Event visualization schema

Dx.3 TCP Trajectory & Geometry Mapping Framework

Duration

2 weeks

Detailed Task

This is one of the most important activities.

The objective is transforming TCP positions into geometry-aware visualization information.

Inputs:

TCP X

TCP Y

TCP Z

Timestamp

Part Geometry

The framework should create:

TCP trajectory paths

Visited regions

Coverage visualization

Position history

The team must define how timestamps are mapped onto geometry.

This allows the HMI to show:

Robot was here

Process anomaly occurred here

KPI variation occurred here

Output

- Geometry mapping engine
 - TCP visualization schema
-

Dx.4 KPI-to-Geometry Superimposition Framework

Duration

1.5 weeks

Detailed Task

The objective is mapping KPI values to physical locations on the part.

Example:

Tension = 35 N
at XYZ location

The framework should support:

Heatmaps
Color overlays
Severity overlays
Region-based visualization

This creates the foundation for:

Process variation visualization
KPI distribution visualization

Output

- KPI overlay framework
-

Dx.5 Anomaly Visualization Framework

Duration

1 week

Detailed Task

The anomaly engine will generate:

Anomaly Timestamp
Severity
Type

This task transforms anomaly outputs into HMI-ready objects.

Examples:

Warning marker

Critical marker

Anomaly timeline event

Anomaly location marker

The framework should define:

Visualization rules

Color rules

Filtering rules

Output

- Anomaly visualization model
-

Dx.6 Expected-vs-Actual Visualization Framework

Duration

1 week

Detailed Task

The System Identification framework produces:

Expected Process Behavior

The visualization layer must prepare:

Expected KPI

Actual KPI

Deviation

Deviation %

The framework should support:

Trend comparison
Deviation plots
Expected-vs-Actual overlays

Output

- Expected-vs-Actual visualization datasets
-

Dx.7 Root-Cause Visualization Framework

Duration

1 week

Detailed Task

The objective is preparing investigation views for engineers.

The framework converts root-cause outputs into:

Possible Cause
Confidence
Related KPIs
Related Events
Affected Geometry Region

This task focuses purely on visualization preparation.

No analytics are performed here.

Output

- Root-cause visualization schema
-

Dx.8 Visualization API & Backend Integration

Duration

1 week

Detailed Task

The final activity integrates all visualization datasets with the HMI backend.

The team develops:

KPI APIs
Geometry APIs
Anomaly APIs
Expected-vs-Actual APIs
Root Cause APIs

The objective is providing visualization-ready payloads for the frontend.

Output

- Visualization service APIs
 - Backend integration complete
-

Recommendation

In your roadmap I would place this deliverable between:

Time-Series Analysis
System Identification
↓
Visualization Analytics Layer
↓
HMI Backend
↓
HMI Frontend

because this layer is what converts analytics outputs into something the HMI can actually display. Without it, the HMI team will receive raw numbers but not visualization-ready information.

Time-Series Analysis System Identification

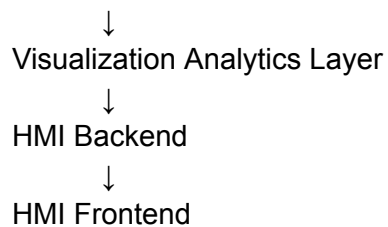
this is will not be ready by this time

we are starting data analyais (acquired from DAQ), TCP actual (nc programmed) before time series and system identification

Exactly. Based on the latest scope and what you described, this deliverable should **not depend on Time-Series Analysis or System Identification**.

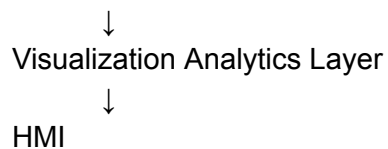
The flow should actually be:

WP5.1 DAQ
(Cleaned, filtered, validated data)

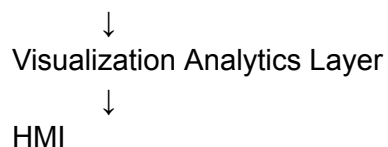


Only later:

Time-Series Anomaly Detection



System Identification



So your first version of visualization can start much earlier.

New Deliverable

Dx – Visualization Data Preparation & Process Mapping Framework

Objective

Transform DAQ outputs into visualization-ready datasets for HMI consumption.

Focus areas:

- KPI visualization
- TCP trajectory visualization
- Process execution visualization
- Process-state visualization
- Geometry mapping
- Historical playback

No anomaly detection.

No behavioral models.

No system identification.

Dx.1 KPI Visualization Dataset Preparation

Duration: 1.5 weeks

Detailed Task

The objective is to transform process measurements into datasets that can be directly consumed by dashboards.

Starting from validated DAQ outputs, the team identifies all KPIs that should appear in the HMI. Examples include tension, speed, temperature, pressure, machine state, material usage and process-health indicators.

For each KPI, the team defines:

- update frequency

- aggregation strategy
- visualization type
- historical retention

The task also creates standardized KPI payload structures so every dashboard uses the same format.

Output

- KPI catalog
 - KPI visualization schema
 - Dashboard-ready KPI payloads
-

Dx.2 Process Timeline & Event Mapping

Duration: 1 week

Detailed Task

The objective is converting raw machine events into process timelines.

Examples:

Course Start
Course End
Pause
Resume
Machine Stop
Operator Action
Alarm

The team develops logic to organize events into chronological process execution timelines.

This becomes the basis for process playback and execution history visualization.

Output

- Event timeline framework
 - Process execution dataset
-

Dx.3 TCP Trajectory Processing

Duration: 2 weeks

Detailed Task

The objective is converting raw TCP positions into visualization-ready trajectories.

Inputs:

Timestamp
TCP Position
TCP Orientation
Machine State

The framework reconstructs:

- actual TCP path
- execution trajectory
- visited regions
- process coverage information

The output should allow an engineer to replay the actual machine movement.

Output

- TCP trajectory datasets
 - Trajectory playback datasets
-

Dx.4 CAD/Geometry Mapping Framework

Duration: 2 weeks

Detailed Task

The objective is establishing the relationship between process data and part geometry.

The framework loads:

- CAD models
- mesh models

- manufacturing geometry

and creates mappings between:

TCP position

Process timestamp

Geometry coordinates

This enables future overlays.

For example:

show where the robot was

show where KPI changed

show where anomaly occurred

Output

- Geometry indexing engine
 - Geometry mapping service
-

Dx.5 KPI-to-Geometry Overlay Framework

Duration: 1.5 weeks

Detailed Task

The objective is projecting process measurements onto the part geometry.

Examples:

Tension along path

Speed along path

Temperature distribution

Pressure distribution

The framework creates visualization-ready objects for:

- heatmaps
- colored path overlays
- process intensity views

No anomaly detection is involved.

Pure visualization.

Output

- KPI overlay datasets
 - Geometry visualization payloads
-

Dx.6 Actual-vs-Planned Execution Visualization

Duration: 1 week

Detailed Task

The objective is comparing:

NC Program
Actual Execution

The framework calculates:

- path deviations
- velocity deviations
- execution timing differences
- skipped regions

These results are prepared for visualization.

Output

- Planned vs actual datasets
 - Deviation visualization datasets
-

Dx.7 Visualization Backend APIs

Duration: 1 week

Detailed Task

The objective is exposing all visualization information through standardized APIs.

The team creates APIs for:

- KPI dashboards
- timelines
- geometry overlays
- trajectory playback
- planned-vs-actual comparison

Output

- Visualization API layer
-

Dx.8 Validation & Demo Package

Duration: 1 week

Detailed Task

The objective is validating the complete visualization pipeline before anomaly detection and behavioral models are available.

The team verifies:

- KPI visualization
- trajectory playback
- geometry overlays
- timeline rendering
- API performance

A demonstrator can already be shown to stakeholders.

Output

- Visualization demonstrator
- Validation report

This deliverable is actually a very good **Ver1 candidate**, because it allows you to show something tangible to management:

DAQ Data



KPI Dashboards

DAQ Data



TCP Path Visualization

DAQ Data



Process Timeline

DAQ Data



3D Geometry Overlay

without waiting for Time-Series Analysis or System Identification to mature. Later, anomaly markers and expected-vs-actual behavioral outputs can simply be added as additional layers on top of the same visualization framework.

can you give as a proper with detailed information about what are planning to do
also keep in mind that we have data set preparation deliverable prior to this

Yes. Since you already have a separate deliverable for:

D3 – Modeling Data Preparation

(where datasets are prepared for:

- Time-Series Analysis

- System Identification
- HMI Visualization

),

then this new deliverable should **not contain data preparation activities again**.

Instead, it should assume:

Validated DAQ Data

+

Prepared Visualization Datasets
(from D3)



Visualization Analytics & Process Mapping Framework



HMI Backend



HMI Frontend

D5 – Visualization Analytics & Process Mapping Framework

Objective

The objective of this deliverable is to transform prepared process datasets into visualization-ready information that can be consumed by the HMI.

This deliverable focuses on creating the intermediate analytics layer between the data platform and the visualization platform.

The framework is responsible for:

- KPI visualization preparation
- Process timeline generation
- TCP trajectory processing
- CAD/geometry mapping
- KPI-to-geometry overlays
- Actual-vs-planned process comparison
- Visualization service APIs

This deliverable does not perform:

- anomaly detection
- system identification
- behavioral modelling

Those outputs will be integrated later.

The purpose is to establish a reusable visualization foundation before advanced analytics become available.

Expected Outputs

Visualization Data Services

Provide structured datasets for:

- KPI dashboards
 - Process timelines
 - TCP trajectories
 - Process-state visualization
 - Geometry overlays
-

Geometry Mapping Engine

Provide mapping between:

Timestamp
TCP Position
Process KPI
Part Geometry

to support future process anomaly and process variation visualization.

Planned-vs-Actual Visualization Framework

Provide datasets enabling:

Programmed path
Actual path
Path deviations
Execution deviations

to be visualized inside HMI.

HMI Backend Integration APIs

Provide APIs supplying visualization-ready information to the HMI backend.

Estimated Duration

Total Duration

8–10 Weeks

D5.1 KPI Visualization Framework

Duration

1.5 Weeks

Detailed Description

The objective of this task is to establish a common framework for visualizing manufacturing KPIs inside the HMI.

The team will review all KPI outputs prepared during the Data Preparation deliverable and identify:

- KPIs requiring visualization
- update frequency
- aggregation methods
- retention requirements
- visualization requirements

For each KPI, a standardized visualization structure will be defined to ensure consistent presentation across all HMI screens.

The framework will define:

Current value
Trend
Historical evolution
Moving average
Process limits
Status indicators

This task creates the foundation for KPI dashboards.

Output

- KPI visualization catalogue
 - KPI visualization schemas
 - Dashboard-ready KPI datasets
-

D5.2 Process Timeline & Event Visualization Framework

Duration

1 Week

Detailed Description

The objective of this task is to transform machine and process events into meaningful process execution timelines.

The team will analyze available process states and machine events and define how they should be represented inside the HMI.

Examples include:

Course Start
Course End
Machine Running
Pause

Resume
Operator Action
Alarm

The framework will establish:

- event categorization
- event severity
- event chronology
- process-state transitions

The generated datasets will support process playback and execution history visualization.

Output

- Event visualization model
 - Process timeline datasets
 - Process-state visualization datasets
-

D5.3 TCP Trajectory Processing Framework

Duration

2 Weeks

Detailed Description

The objective of this task is to transform recorded TCP positions into visualization-ready process trajectories.

The team will process:

TCP Position
TCP Orientation
Timestamp
Machine State

and reconstruct:

- actual machine trajectories

- execution paths
- process progression

The framework will support trajectory replay and execution analysis.

The resulting datasets will later be visualized on top of the part geometry.

Output

- TCP trajectory datasets
 - Process replay datasets
 - Trajectory visualization services
-

D5.4 Geometry Mapping Framework

Duration

2 Weeks

Detailed Description

The objective of this task is to establish the relationship between process data and physical part geometry.

The team will develop services capable of loading:

CAD Geometry
Mesh Models
Part Representations

and mapping process information onto those geometries.

The framework will define relationships between:

Timestamp
TCP Position
Part Coordinates
Geometry Regions

This creates the foundation required for future anomaly overlays and process variation visualization.

Output

- Geometry indexing framework
 - Spatial mapping services
 - Geometry lookup engine
-

D5.5 KPI-to-Geometry Visualization Framework

Duration

1.5 Weeks

Detailed Description

The objective of this task is to project KPI values onto the corresponding regions of the part geometry.

The framework will convert process measurements into visualization-ready overlays.

Examples include:

Tension distribution

Speed distribution

Temperature distribution

Pressure distribution

The team will define methods for:

- heatmap generation
- colored overlays
- region-based highlighting
- process intensity visualization

This task focuses solely on visualization preparation and not anomaly detection.

Output

- KPI overlay framework
- Geometry overlay datasets
- Visualization-ready KPI mapping services

D5.6 Actual-vs-Planned Visualization Framework

Duration

1 Week

Detailed Description

The objective of this task is to prepare datasets enabling comparison between programmed execution and actual execution.

Inputs include:

NC Program
Actual TCP Trajectory
Execution Logs

The framework will generate visualization information describing:

- path deviations
- timing deviations
- velocity differences
- execution mismatches

The output will support future engineering investigations inside the HMI.

Output

- Actual-vs-planned datasets
- Deviation visualization services

D5.7 Visualization Backend Integration APIs

Duration

1 Week

Detailed Description

The objective of this task is to expose all visualization information through reusable backend services.

The team will define and implement APIs for:

- KPI visualization
- Process timelines
- TCP trajectories
- Geometry overlays
- Planned-vs-actual comparisons

These APIs will become the interface between the Visualization Analytics Layer and the HMI Backend.

Output

- Visualization API specification
 - Backend integration services
-

D5.8 Validation & Demonstration

Duration

1 Week

Detailed Description

The objective of this task is to validate the complete visualization pipeline before anomaly detection and system identification modules become available.

The team will verify:

- KPI visualization correctness
- TCP replay functionality
- Geometry overlay functionality
- Timeline rendering
- API performance

A demonstration package will be prepared to showcase the complete visualization workflow.

Output

- Visualization demonstrator
- Validation report
- Release package for HMI integration

This deliverable fits naturally **after D3 (Data Preparation)** and **before Time-Series Analysis and System Identification**, allowing WP5.3 to start delivering visible functionality without waiting for the analytics models to mature.

example , i am expecting like this:

API Development

Duration: 2 weeks

Activities:

Dashboard APIs

Anomaly APIs

Root-cause APIs

Geometry APIs

Historical query APIs

what are the activities or tasks we need to plan in each sub deliverables

example , i am expecting like this:

API Development

Duration: 2 weeks

Activities:

Dashboard APIs

Anomaly APIs

Root-cause APIs

Geometry APIs

Historical query APIs

what are the activities or tasks we need to plan in each sub deliverables

Got it. You are not asking for the **description of the deliverable**, but rather the **actual engineering tasks/activities that need to be planned under each sub-deliverable**, similar to:

API Development
Duration: 2 weeks

Activities:

- Dashboard APIs
- Anomaly APIs
- Root-cause APIs
- Geometry APIs
- Historical query APIs

For your **Visualization Analytics & Process Mapping Framework**, I would define the activities like this:

D5.1 KPI Visualization Framework

Duration: 1.5 Weeks

Activities

KPI Identification & Classification

- Identify KPIs available from DAQ
- Categorize KPIs into process KPIs, machine KPIs and quality KPIs
- Define KPI ownership and source

KPI Aggregation Logic

- Define instantaneous KPIs
- Define moving-average KPIs
- Define aggregated KPIs
- Define KPI calculation windows

KPI Visualization Model Definition

- Define card views
- Define trend chart views
- Define gauge views
- Define status indicator views

KPI Data Structures

- Define KPI payload schema
- Define timestamp handling
- Define KPI metadata structure

KPI Validation

- Verify KPI consistency
 - Validate KPI calculations
 - Validate visualization readiness
-

D5.2 Process Timeline & Event Visualization Framework

Duration: 1 Week

Activities

Process Event Identification

- Identify machine events
- Identify process events
- Identify operator events

Event Categorization

- Normal events
- Warning events
- Critical events
- Informational events

Timeline Construction

- Create process timeline structure
- Define event ordering logic

- Define event grouping logic

Process-State Modeling

- Define process states
- Define state transitions
- Define state duration calculations

Timeline Dataset Validation

- Verify event sequencing
 - Verify state transitions
 - Validate timeline visualization datasets
-

D5.3 TCP Trajectory Processing Framework

Duration: 2 Weeks

Activities

TCP Data Acquisition Interface

- Import TCP position data
- Import TCP orientation data
- Import execution timestamps

Trajectory Reconstruction

- Generate actual TCP path
- Generate trajectory segments
- Generate process path history

Trajectory Optimization

- Remove duplicate points
- Interpolate missing points
- Smooth trajectory if required

Process Replay Dataset Generation

- Create playback datasets

- Create timestamp-indexed trajectory datasets
- Create trajectory snapshots

Validation

- Validate trajectory continuity
 - Validate path reconstruction
 - Compare against actual execution logs
-

D5.4 Geometry Mapping Framework

Duration: 2 Weeks

Activities

Geometry Import Services

- CAD model loading
- Mesh loading
- Geometry metadata extraction

Geometry Indexing

- Create geometry regions
- Create spatial indexing structures
- Create lookup tables

TCP-to-Geometry Mapping

- Map TCP positions to geometry
- Map trajectory segments to geometry regions
- Create geometry references

Region Definition

- Define geometric zones
- Define process regions
- Define visualization regions

Validation

- Validate mapping accuracy
- Validate geometry indexing

- Validate region assignment
-

D5.5 KPI-to-Geometry Overlay Framework

Duration: 1.5 Weeks

Activities

KPI Mapping Logic

- Map KPI values to TCP positions
- Map KPI values to geometry regions
- Associate KPI timestamps with geometry

Overlay Generation

- Generate heatmaps
- Generate color overlays
- Generate path overlays

Visualization Rules

- Define color scales
- Define threshold ranges
- Define severity representation

Overlay Data Services

- Generate overlay payloads
- Generate region-level summaries
- Generate overlay metadata

Validation

- Validate overlay positioning
 - Validate KPI-to-geometry mapping
 - Validate visualization consistency
-

D5.6 Actual-vs-Planned Visualization Framework

Duration: 1 Week

Activities

Planned Path Processing

- Import NC path
- Extract programmed trajectory
- Extract programmed timestamps

Actual Path Processing

- Import actual execution path
- Extract execution trajectory
- Extract actual timestamps

Path Comparison

- Compute path deviations
- Compute timing deviations
- Compute velocity deviations

Visualization Dataset Generation

- Generate deviation markers
- Generate comparison overlays
- Generate comparison metrics

Validation

- Verify comparison logic
- Verify deviation calculations
- Validate visualization datasets

D5.7 Visualization Backend APIs

Duration: 1 Week

Activities

Dashboard APIs

- KPI dashboard APIs
- KPI summary APIs
- Trend APIs

Timeline APIs

- Event timeline APIs
- Process-state APIs
- Event history APIs

Trajectory APIs

- TCP trajectory APIs
- Playback APIs
- Position query APIs

Geometry APIs

- Geometry loading APIs
- Overlay APIs
- Region query APIs

Comparison APIs

- Planned-vs-actual APIs
- Deviation APIs
- Historical comparison APIs

API Validation

- Response validation
- Performance testing
- Integration testing

D5.8 Validation & Demonstration

Duration: 1 Week

Activities

Functional Validation

- KPI visualization validation
- Timeline validation
- Trajectory validation
- Geometry overlay validation

Integration Validation

- Data layer integration
- Backend integration
- Frontend integration

Performance Validation

- API performance testing
- Rendering performance testing
- Dataset loading validation

Demonstration Package Preparation

- Demo scenarios
- Demo datasets
- Demo scripts
- User walkthrough

Final Review

- Stakeholder review
- Feedback incorporation
- Release sign-off

This level of detail is usually what project managers and work-package leads expect when estimating effort and assigning ownership to individual engineers.

we don't have to do any KPI calculation or aggregation , this is done by work package 5.1

Good catch. If WP5.1 is responsible for:

- Data acquisition
- Data cleaning
- Data validation
- Data preprocessing

- KPI calculation
- KPI aggregation

then WP5.3 should **not duplicate any of those activities**.

In that case, D5 should focus purely on:

Visualization preparation
Geometry mapping
Process mapping
Trajectory visualization
Timeline generation
Visualization APIs

rather than KPI engineering.

D5.1 KPI Visualization Framework

Duration: ~1 Week

Activities

KPI Inventory Review

- Review KPIs provided by WP5.1
- Review KPI metadata
- Review KPI update frequency
- Review KPI ownership and source

KPI Visualization Requirements

- Define which KPIs appear in which HMI screens
- Define dashboard layouts
- Define chart types
- Define KPI grouping strategy

KPI Visualization Data Structures

- Define frontend payload structure
- Define visualization metadata
- Define KPI status representation

KPI Visualization Validation

- Verify KPI availability
- Verify KPI payload compatibility
- Verify dashboard readiness

Output

- KPI visualization specification
 - KPI display model
 - HMI KPI mapping document
-

D5.2 Process Timeline & Event Visualization Framework

Duration: ~1 Week

Activities

Process Event Identification

- Review available machine events
- Review available process events
- Review available alarm events

Event Categorization

- Define event classes
- Define event severity categories
- Define event display priorities

Timeline Representation

- Define timeline structures
- Define process-state visualization
- Define event visualization logic

Timeline Dataset Validation

- Verify event sequence correctness
- Verify process-state transitions

Output

- Timeline visualization model
 - Event visualization datasets
-

D5.3 TCP Trajectory Processing Framework

Duration: ~2 Weeks

Activities

TCP Data Interface Definition

- Review TCP data structure
- Review timestamp synchronization
- Review coordinate systems

Trajectory Reconstruction

- Generate actual TCP trajectories
- Create trajectory segments
- Generate trajectory history

Trajectory Playback Dataset Generation

- Generate replay datasets
- Generate timestamp-indexed trajectory datasets
- Generate playback metadata

Trajectory Validation

- Validate path continuity
- Validate coordinate consistency
- Validate playback correctness

Output

- TCP trajectory datasets
 - Process replay datasets
-

D5.4 Geometry Mapping Framework

Duration: ~2 Weeks

Activities

Geometry Import Services

- Load CAD/mesh models
- Extract geometry metadata
- Create geometry references

Geometry Indexing

- Create spatial indexing structures
- Define geometry regions
- Define visualization zones

TCP-to-Geometry Mapping

- Map TCP positions to geometry
- Map trajectory segments to geometry regions
- Create geometry references

Geometry Validation

- Validate mapping accuracy
- Validate geometry indexing
- Validate region assignment

Output

- Geometry mapping framework
 - Spatial indexing services
-

D5.5 KPI-to-Geometry Overlay Framework

Duration: ~1.5 Weeks

Activities

KPI-to-Position Association

- Associate KPIs with TCP positions
- Associate KPIs with timestamps
- Associate KPIs with geometry regions

Overlay Generation Logic

- Generate heatmap datasets
- Generate color overlay datasets
- Generate path-based overlay datasets

Visualization Rule Definition

- Define color schemes
- Define overlay rules
- Define visualization thresholds

Overlay Validation

- Validate KPI location mapping
- Validate overlay generation
- Validate visualization consistency

Output

- KPI overlay datasets
 - Geometry overlay services
-

D5.6 Planned-vs-Actual Visualization Framework

Duration: ~1 Week

Activities

Planned Path Import

- Import NC path information
- Import planned trajectory
- Import planned timestamps

Actual Path Import

- Import actual execution trajectory
- Import actual timestamps

Path Comparison Logic

- Calculate positional deviations
- Calculate timing deviations
- Calculate execution differences

Visualization Dataset Generation

- Generate deviation datasets
- Generate comparison overlays
- Generate visualization markers

Output

- Planned-vs-actual datasets
 - Deviation visualization services
-

D5.7 Visualization Backend APIs

Duration: ~1 Week

Activities

Dashboard APIs

- KPI dashboard endpoints
- KPI history endpoints
- KPI visualization endpoints

Timeline APIs

- Event timeline endpoints
- Process-state endpoints
- Event history endpoints

Trajectory APIs

- TCP trajectory endpoints

- Replay endpoints
- Position query endpoints

Geometry APIs

- Geometry retrieval endpoints
- Overlay endpoints
- Region query endpoints

Comparison APIs

- Planned-vs-actual endpoints
- Deviation endpoints

API Validation

- API testing
- Performance testing
- Integration testing

Output

- Visualization API layer
-

D5.8 Validation & Demonstration

Duration: ~1 Week

Activities

Functional Validation

- KPI visualization validation
- Timeline validation
- Trajectory validation
- Geometry overlay validation

Integration Validation

- Visualization layer ↔ HMI backend validation
- API validation
- Frontend integration validation

Demonstration Preparation

- Demo datasets
- Demo scenarios
- Demo scripts
- User walkthrough preparation

Output

- Visualization demonstrator
- Validation report

This version is much more aligned with the latest WP5.3 scope because **WP5.3 is consuming and visualizing KPIs from WP5.1**, not calculating them. The focus is on **mapping, contextualizing, and visualizing data**, especially on timelines, trajectories, and 3D geometry.

now can you give the list of HMI deliverables and associated with it?

Based on everything we've discussed, I would structure the **HMI workstream** as a separate set of deliverables under WP5.3.

The important distinction is:

Visualization Analytics Layer



HMI Backend



HMI Frontend



HMI Releases (Ver1, Ver2, Ver3)

So I would recommend the following.

H1 — HMI Architecture & Visualization Framework Design

Duration

~4–5 Weeks

Objective

Define the overall HMI architecture, visualization strategy, user workflows, frontend/backend architecture, API contracts, and data models.

Key Tasks

H1.1 Requirements Analysis

- Define process engineer use cases
- Define manufacturing engineer use cases
- Define anomaly visualization requirements
- Define KPI visualization requirements
- Define expected-vs-actual visualization requirements
- Define geometry overlay requirements

H1.2 HMI Architecture Definition

- Frontend architecture
- Backend architecture
- Deployment architecture
- API architecture
- Security architecture

H1.3 Visualization Design

- KPI dashboard wireframes
- Geometry visualization wireframes
- Timeline visualization wireframes
- Investigation dashboard wireframes

H1.4 Data Model & Interface Definition

- KPI schemas
- Geometry schemas
- Timeline schemas

- API payload definitions

Outputs

- HMI Architecture Document
 - UI Wireframes
 - API Specifications
 - Data Model Specifications
-

H2 — Visualization Analytics & Process Mapping Framework

Duration

~8–10 Weeks

Objective

Convert DAQ data and process execution data into visualization-ready information for HMI consumption.

This deliverable starts immediately after Data Preparation and does not depend on Time-Series Analysis or System Identification.

Key Tasks

H2.1 KPI Visualization Framework

- KPI inventory review
- KPI display model definition
- KPI visualization structures
- Dashboard mapping

H2.2 Process Timeline & Event Visualization

- Event categorization
- Process-state representation
- Timeline generation
- Event sequencing

H2.3 TCP Trajectory Processing

- TCP path reconstruction
- Trajectory playback datasets
- Path history generation
- Execution replay preparation

H2.4 Geometry Mapping Framework

- CAD loading
- Geometry indexing
- TCP-to-geometry mapping
- Region definition

H2.5 KPI-to-Geometry Overlay Framework

- KPI-to-position mapping
- Heatmap generation
- Overlay dataset generation

H2.6 Planned-vs-Actual Visualization Framework

- NC path import
- Actual path import
- Path comparison
- Deviation dataset generation

H2.7 Visualization APIs

- Dashboard APIs
- Timeline APIs
- Geometry APIs
- Trajectory APIs
- Comparison APIs

H2.8 Validation & Demonstration

- Functional validation
- Integration validation
- Demonstration preparation

Outputs

- Visualization Analytics Layer
- Geometry Mapping Engine
- Visualization APIs
- Demonstration Package

H3 — HMI Backend Development

Duration

~6–8 Weeks

Objective

Develop backend services that expose visualization-ready information to frontend applications.

Key Tasks

H3.1 Backend Framework Setup

- Backend project setup
- Database setup
- Logging
- Deployment setup

H3.2 Data Integration Services

- Visualization dataset ingestion
- Metadata ingestion
- Dataset synchronization

H3.3 Geometry Services

- Geometry retrieval services
- Overlay services
- Region query services

H3.4 Analytics Integration Services

- Time-series integration
- Behavioral-model integration
- Expected-vs-actual integration

H3.5 API Development

- Dashboard APIs
- Geometry APIs
- Timeline APIs

- Historical query APIs

H3.6 Backend Validation

- Performance testing
- API testing
- Integration testing

Outputs

- Backend Services
 - API Layer
 - Geometry Services
-

H4 — HMI Frontend Development (Ver1)

Duration

~4–5 Weeks

Objective

Release the first HMI version demonstrating the framework and connectivity.

Scope

This version does **not** depend on mature anomaly detection or system identification outputs.

Key Tasks

H4.1 UI Framework Setup

- Navigation structure
- Layout framework
- Theme implementation

H4.2 Dashboard Development

- KPI placeholders
- Basic charts
- Static views

H4.3 Frontend–Backend Integration

- API integration
- Data loading
- State management

H4.4 Basic Geometry Viewer

- Geometry loading
- Navigation
- Object selection

H4.5 Testing & Release

- Navigation testing
- API testing
- Demo preparation

Outputs

- HMI Ver1
 - Navigation Framework
 - Basic Dashboards
 - Backend Connectivity
-

H5 — HMI Enhancement & Analytics Integration (Ver2)

Duration

~6–8 Weeks

Objective

Integrate actual analytics outputs and production data into the HMI.

Key Tasks

H5.1 Production Data Integration

- WP5.1 integration

- Actual process data integration

H5.2 Anomaly Visualization

- Time-series anomaly visualization
- Severity visualization
- Anomaly timeline

H5.3 Geometry Overlays

- KPI overlays
- Anomaly overlays
- Process variation overlays

H5.4 Investigation Views

- Anomaly investigation
- Historical analysis
- Trend analysis

H5.5 Validation

- Production data validation
- Visualization validation

Outputs

- HMI Ver2
- Anomaly Dashboards
- Geometry Overlays
- Investigation Views

H6 — HMI Process Intelligence Platform (Ver3)

Duration

~6–8 Weeks

Objective

Deliver the final HMI integrating all WP5.3 capabilities.

Key Tasks

H6.1 Behavioral Model Visualization

- Expected-vs-actual visualization
- Behavioral model outputs
- Deviation visualization

H6.2 Advanced Investigation Workspace

- Process intelligence views
- Historical replay
- Multi-run comparison

H6.3 Root-Cause Analysis Visualization

- Root-cause visualization
- Correlation views
- Investigation workflows

H6.4 Performance Optimization

- Large dataset handling
- Rendering optimization
- Query optimization

H6.5 Final Acceptance Testing

- End-to-end validation
- User acceptance testing

Outputs

- HMI Ver3
- Full Process Intelligence Dashboard
- Engineering Investigation Platform

H7 — Final Deployment & Documentation

Duration

~2–3 Weeks

Key Tasks

- Deployment preparation
- User documentation
- Administrator documentation
- Training material
- Final rollout support

Outputs

- Production Deployment Package
 - User Guide
 - Maintenance Guide
-

Final HMI Roadmap



This structure aligns with your latest WP5.3 scope where the HMI is focused on:

- KPI visualization
- TCP trajectory visualization
- Geometry overlays
- Process anomaly visualization
- Expected-vs-actual analysis
- Root-cause investigation support

and **not** on process control, optimization, or defect prediction.